5TH GRADE MATH: EQUIVALENT FRACTIONS

LESSON & WORKSHEET

WHAT ARE EQUIVALENT FRACTIONS?

Definition: Equivalent fractions are fractions that name the **exact same amount** of a whole, even though they use different numerators and denominators.

1. The Pizza Model (Visual Proof) ■

Imagine you and a friend order two identical pizzas:
• **Pizza A** is cut into **2 equal slices**. You eat 1 slice (1/2).
• **Pizza B** is cut into **4 equal slices**. Your friend eats 2 slices (2/4).

PIZZA A (1/2)

1 out of 2 equal slices

PIZZA B (2/4)

2 out of 4 equal slices

■ **Look at the pizzas:** Both of you ate the **exact same amount** of pizza! Therefore, **1/2 = 2/4**.

2. The Calculator Check (Decimal Verification) ■

You can confirm if fractions are equivalent by converting them into decimals on a calculator. Divide the **numerator (top number)** by the **denominator (bottom number)**:

Fraction	Calculator Division	Decimal Value	Are they equivalent?
1 / 2	1 ÷ 2	0.5	YES ✓
2 / 4	2 ÷ 4	0.5	YES ✓
3 / 6	3 ÷ 6	0.5	YES ✓

■ **Rule:** Because every division result equals **0.5**, all three fractions are **100% equivalent**!

3. How to Create and Test Equivalent Fractions ■

MATH RULES FOR EQUIVALENT FRACTIONS

Multiplying / Dividing Rule: Multiply or divide both numerator and denominator by the same non-zero number:
1/2 = (1 × 2) / (2 × 2) = **2/4**

Cross-Product Check Rule: Two fractions **a/b** and **c/d** are equivalent if **a × d = b × c**:
1/2 = 2/4 because **1 × 4 = 4** and **2 × 2 = 4**.

✏■ Student Activity: Equivalent Fraction Sorting Challenge

Instructions:

1. Simplify each fraction in the Fraction Bank or use a calculator to find its decimal value.
2. Find the equivalent fractions for each target group.
3. Write each fraction into its designated Set Box below!

■ FRACTION BANK (Choose from these 6 fractions):

1 / 3 2 / 5 2 / 6 4 / 10 3 / 9 6 / 15

■ SET 1: Fractions Equivalent to 1/3 (Decimal Value: 0.333...)

1. _____
2. _____
3. _____

■ SET 2: Fractions Equivalent to 2/5 (Decimal Value: 0.4)

1. _____
2. _____
3. _____

TEACHER ANSWER KEY

■ Teacher Answer Key & Explanations

- **Set 1 (Equivalent to 1/3 = 0.333...): 1/3**, **2/6** (2÷2 / 6÷2), **3/9** (3÷3 / 9÷3)
- **Set 2 (Equivalent to 2/5 = 0.4): 2/5**, **4/10** (4÷2 / 10÷2), **6/15** (6÷3 / 15÷3)